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Unit “ 4 : Session “ 5 : SLO “ 1 | Multiple Bus Organization “ Fill in the Blanks

■ I. Fill in the Blanks

1. address bus “ carries memory addresses from the processor to memory.
2. data bus “ transfers data between the processor and I/O devices.
3. Harvard “ separate buses for instructions and data define the Harvard architecture.
4. local (backside/cache) bus “ connects the CPU to high-speed cache memory.
5. contention (bus bottlenecks) “ multiple buses allow simultaneous transfers, reducing contention.
6. Bus A: first source operand to the ALU; Bus B: second source operand (Bus C carries the ALU result back to the registers).
7. control bus “ transmits control signals between components.
8. performance (throughput) “ parallelism across buses improves overall speed over a single-bus system.
9. bus arbiter “ coordinates traffic and grants access among devices/buses.
10. crossbar (point-to-point interconnect) “ modern processors use high-speed crossbar/point-to-point buses to link multiple cores to the shared cache.